Design challenges

In the Indian market, price is a huge challenge. Besides adding new features, one needs to get the price right. And that means getting the technology right.

"The better your technology, the less hardware you need," says Nikhil Bhaskaran, founder of ShunyaOS. "And the less hardware you need, the less expensive your product becomes."

With respect to the operating system, standardisation needs to take place both on microprocessor and microcontroller fronts.

Then there is the question of optimum power consumption for longer battery life. This becomes even more pertinent as the next-gen devices are going to be slimmer and packed with many features that will require intense processing power.

"We need to work towards making such devices more and more intuitively smarter. Although we have access to the required hardware, there is a shortage of design houses that can power pack all the components to deliver the right kind of output," says Amit Khatri, co-founder, Noise.

"Hardware, software, and post-data processing are the three areas to be worked on."

Lack of ecosystem

As Nikhil Bhaskaran points out, large scale collaborations need to take place rather than working individually on projects related to electronic designing.

"There is a lack of an ecosystem having no collaboration. If we are able to develop a platform, where knowledge sharing happens, then it will give lots of opportunities to people and also reduce costs of procuring materials from abroad," says Aashish Kumbhat.

Recounting the earlier days of starting his company, Pratik Saraogi, CEO of Actifit, says they were filled with lots of struggle—from acquiring suitable PCBs to months of product development. All this makes the scarcity of large-scale manufacturing all the more glaring.

Global market challenges

Closely related to the above is the issue of developing exceptional prototypes and then trying to compete with companies based in the US, Singapore, etc. Companies in these countries are already developing quality products. So if you do not have high-skill prototypes, it is challenging to sell the finished products.

"We need to have a base ecosystem that allows you to reach a high-fidelity prototype," says Abhishek Satish, co-founder and director of Vicara.

All the above challenges boil down to the fact that large scale collaborations, involving people from both the tech and business sides, do not happen. Tech needs to follow business and not the other way around.

What can help?

It is really helpful if one can have a qualified contract manufacturer to oversee the coordination between you and the manufacturing process.

Another news of respite is that the smart device market will continue to grow. Even if it takes many years, eventually the wearable space will overtake the smartphone market. To reach that level, people, particularly Indians, should focus on capturing mass quantity to generate enough revenue. This in turn will enable the production of high-class products at low cost.

"As long as there is hardware involved, volumes matter a lot," says Nikhil Bhaskaran.

Man wearing headphones and smart glasses mockup (Image from rawpixel)

Speaking on the same subject, Pratik Saraogi maintains that besides hardware and software, wearables also incorporate firmware that includes stringent designing and complex processing. And it seems a mass market exists for people who are actively into fitness and exercising. To leverage this, smart wearables can be incorporated with better algorithms having functional designs that fulfil various requirements.

From a tech perspective, a huge amount of integration between different technologies will happen. Soon a camera-enabled microprocessor will be available that will be able to perform the functions of a sensor.

New set of emerging wearables

The wearable market is much beyond Fitbits, as hearables and eyewear also belong to the family. In future, besides providing data, these devices will control all aspects of our life. Therefore, truly wireless devices have to be enabled with AI. To achieve that, there has to be a need for working on a common stack/chipset and then build use cases around it. Also to mention are human-machine interface (HMI) devices in the industry.

The MTK2503 chipset that supports interactive features for GPS movement,

Expert tips on a proper design approach

1. Develop products while maintaining cost-effectiveness.
2. Analyse the existing competition and make decisions accordingly.
3. Identify the design constraints based on consumer feedback for incorporating hardware and OS with better features.
4. Make sure to serve the underserved consumer needs.
5. The market is wide enough. Therefore, don't limit yourself to a single design-thinking. Be creative and explore several options.
6. Last but important, always design for mass manufacturing.